

Fecha:
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Presentation

EMOTIONAL ASSISTANCE

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INTRODUCTION

1. Facial emotion recognition
 - a. VGG-like with FER2013
 - b. Custom VGG13 with FER2013+
2. Emotional assistance with a chatbot based on therapist conversations.
 - a. Naïve Bayes
 - b. Fine-tuned Distill RoBERTa
 - c. Llama-3.2-1b-instruct

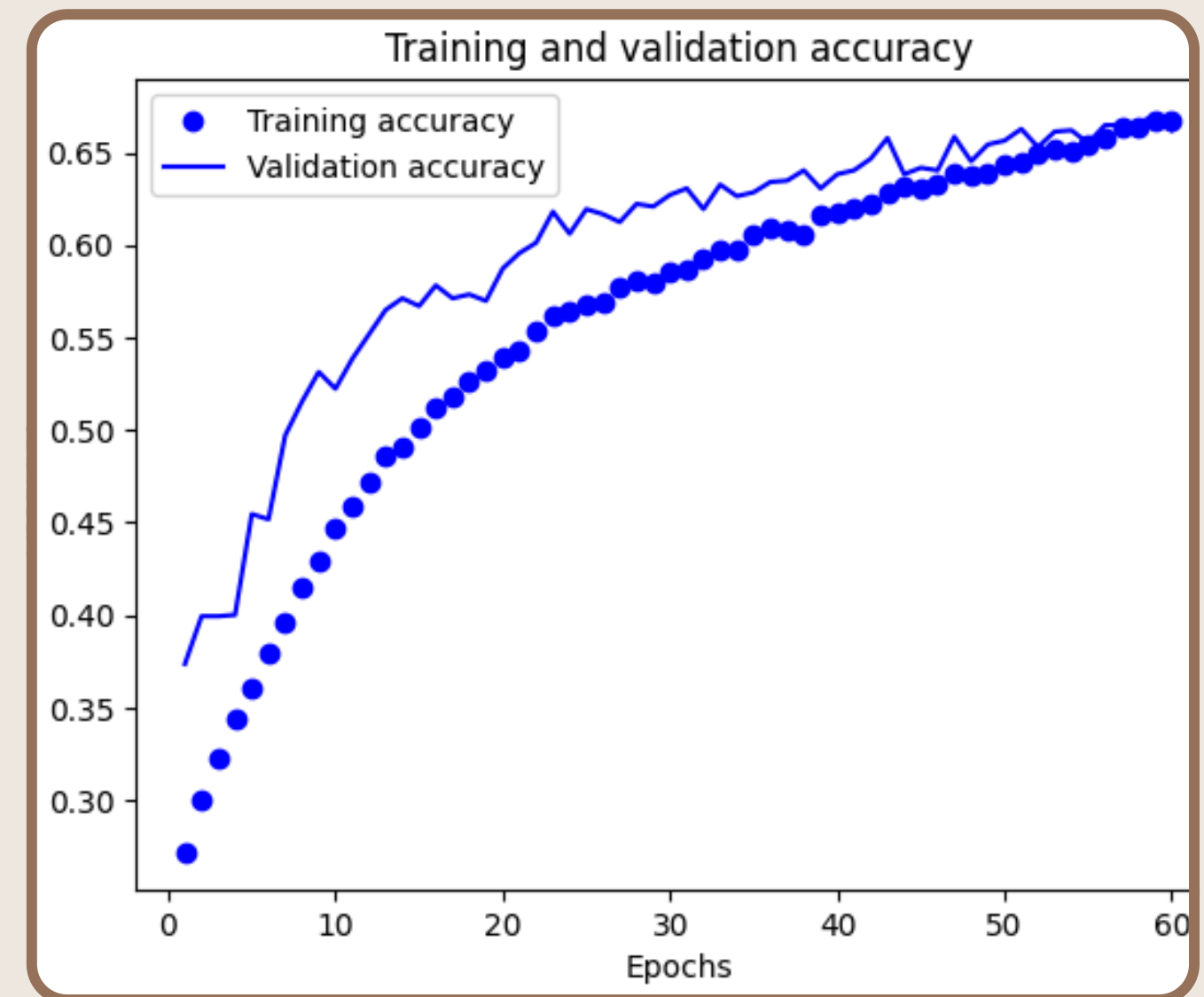
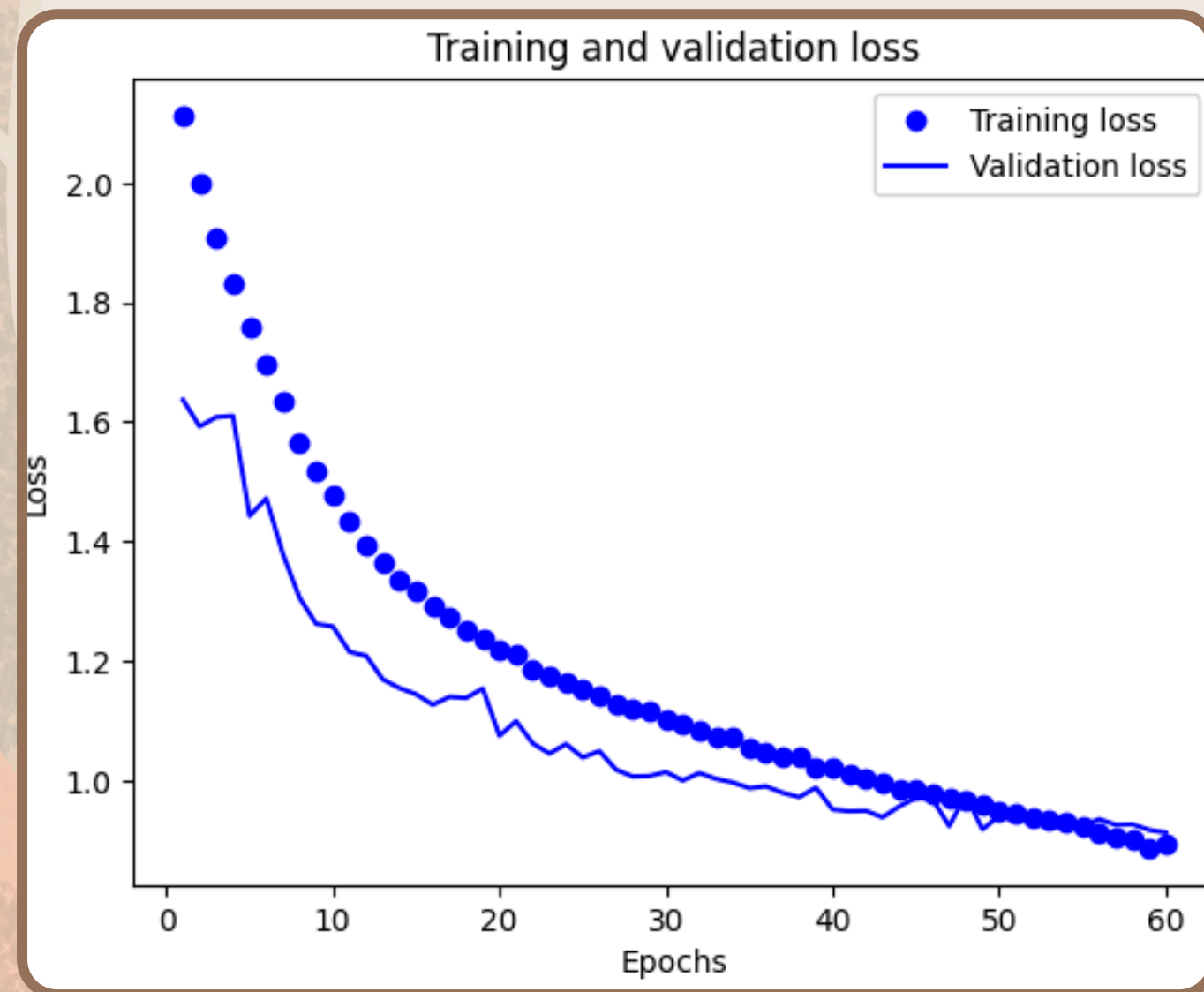
VGG-LIKE WITH FER2013



- Goal: Classify facial expressions into one of six categories: Angry, Fear, Happy, Sad, Surprise, and Neutral.
- Process: using the haarcascade frontalface default to find the faces and classify the emotion with the trained CNN.
- Data: 24,400 48x48 pixel grayscale images of centred faces.
- Preprocessing: data generators and data augmentation.

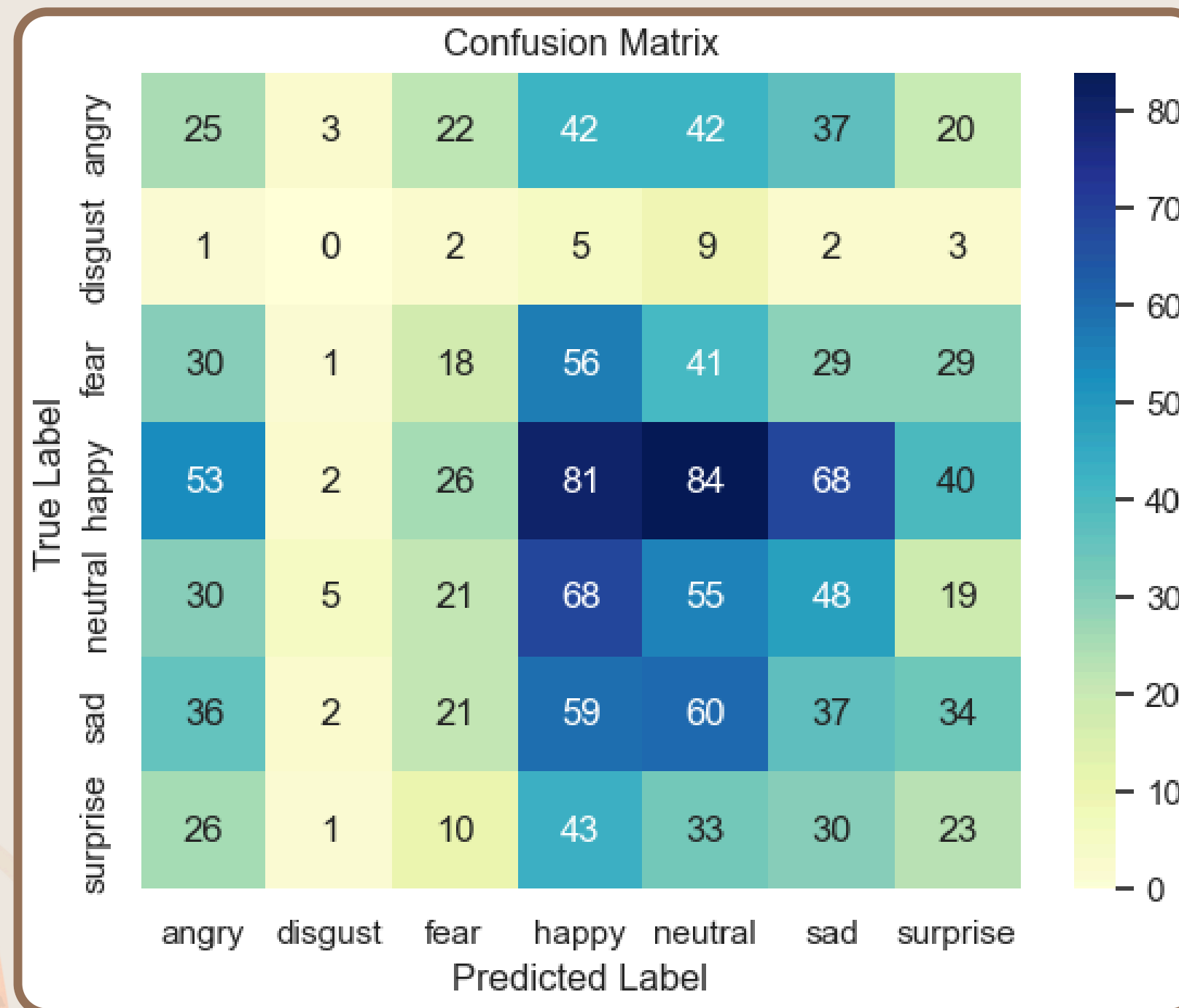
VGG-LIKE WITH FER2013

Evaluation



VGG-LIKE WITH FER2013

Evaluation



CUSTOM VGG₁₃ WITH FER₂₀₁₃⁺

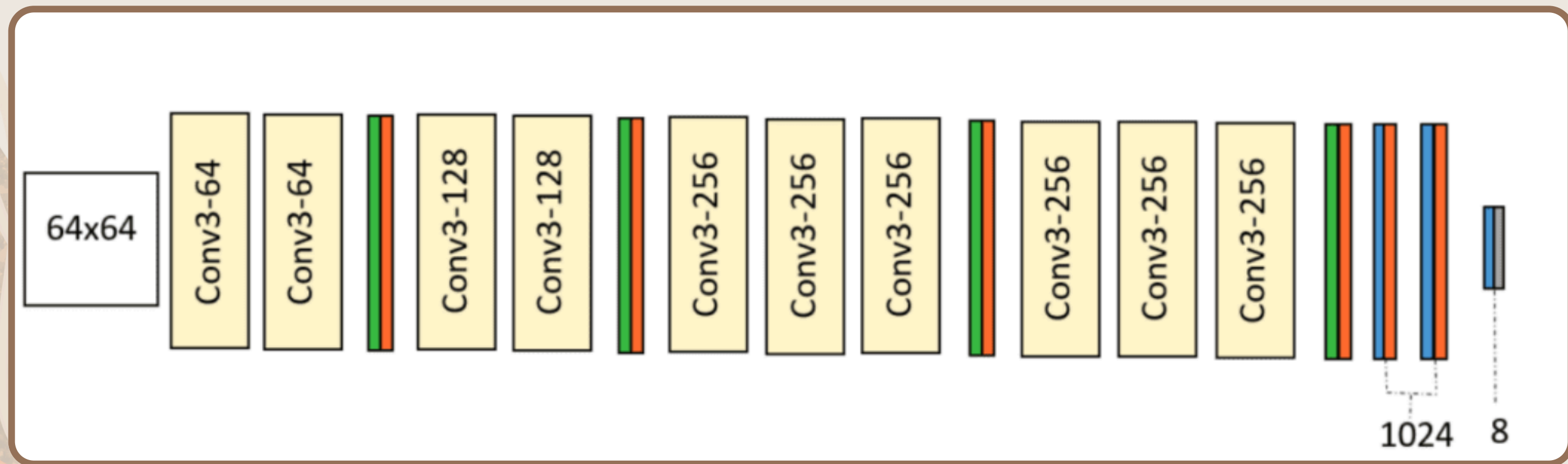
- Face detection: first detect the faces with Ultra-lightweight Face Detection RFB-320 trained with the WIDE FACE dataset that has 32.203 images with 393.703 faces in different positions.

CUSTOM VGG₁₃ WITH FER₂₀₁₃⁺

- Classify emotions: I used the pretrained weights by a custom VGG13 in ONNX format with FER2013+ dataset.
- Data: FER+ dataset is a notable extension of the original Facial Expression Recognition improve upon the limitations.

CUSTOM VGG₁₃ WITH FER₂₀₁₃⁺

- Structure:



MULTINOMIAL NAIVE BAYES

- The first approach is a Multinomial Naive Bayes classifier, which is well-suited for classification tasks involving discrete features, such as word counts in text classification.
- Depending on the user input, it generates a predefined response.
- This approach is very simple and low computationally cost but is impossible to maintain a conversation because is for emotion classification and the meaning of the tokens doesn't depend on the context.

FINE-TUNED DISTILLROBERTA

- Secondly, I realized a pretrained Fine-tuned DistillRoBERTa with a Flask interface.
- The BERT-like models aren't good at all for a chatbot because they process the input bidirectional, they consider the full context of a sentence.
- The BERT-like models are normally use for text classification.
- In my implementation, it classifies the emotion of the text and it output a response in base of the emotion.

LLAMA-3.2-1B-INSTRUCT

- For a chatbot, the best approach is a decoder LLM because it generates text step-by-step predicting the probabilities of the next word.
1. Download the model in LM Studio and call it in a python script.
 2. Interface with Streamlit.
 3. Add a system prompt for putting a therapist specializing context.

NEXT STEPS

- Do a fine-tuned of the LLM.
- Do a logic agent to recommend the user.
- Combine the FER, the chatbot and the logic agent.



THANK